

Qatar-2b

R-1992

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-2_160618 · created 2026-08-07T15:11:36+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457557.738 +/- 0.00029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.226 +/- 0.064
Transit depth (Rp/Rs)^2	5.12 +/- 2.88 [%]
Orbital Inclination (inc)	84.08 +/- 1.77
Airmass coefficient 1 (a1)	0.28 +/- 0.14
Airmass coefficient 2 (a2)	0.85 +/- 0.53
Scatter in the residuals of the lightcurve fit is	52.22 %
Best Comparison Star	None
Optimal Aperture	1.46
Optimal Annulus	3.0
Transit Duration (day)	0.067 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.226 $\pm$ 0.064 vs 0.16327 (deviation 0.76 σ)
Duration fit vs published	0.067 d vs 0.0767 d (deviation 0.62 σ)
Mid-time O-C	0.5 min
Depth SNR	2.7
Residual scatter	52.22 %

Light curve

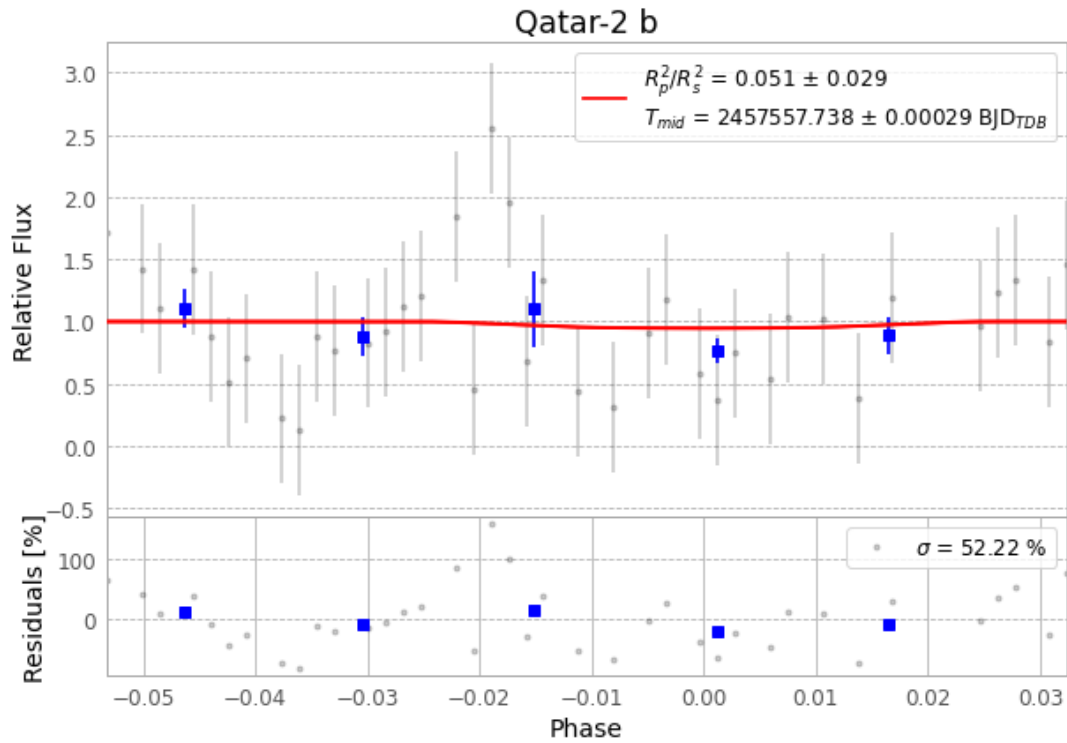


Figure 1 — FinalLightCurve_Qatar-2 b_2016-06-17.png (SHA-256 f18ba04fd4950a46...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [221, 440], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 a20ef5a45f72b678...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

58 FITS frames (38 MB), Qatar-2160618034514.FITS → Qatar-2160618064212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-2-160618.log (64ccc2dfb076...), FinalLightCurve_Qatar-2 b_2016-06-17.png (f18ba04fd495...), FinalLightCurve_Qatar-2 b_2016-06-17.csv (d37f4a3bcc82...)

Compute resources

Wall time 105.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 15:11Z.